Shuqi Ke

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Employment

Shenzhen Shita Robotics Technology Co., Ltd. *Shenzhen, China*
Algorithm Engineer Intern *Feb. 2026 - Aug. 2026*

- Built a real-robot visual-tactile data collection, recording, training, and inference pipeline for embodied intelligence and edge vision; integrated Piper leader-follower control, LTeach tactile cameras, and multimodal data management; adapted ACT and PI0 / PI0.5 training configurations and effective batch-size accounting.
- Developed two-frame motion-vector prediction, multi-camera detection, and template tracking; optimized models and losses; implemented FP16 / INT8 quantization, calibration-set construction, and edge-device consistency validation across PC, RV1126B, and RK3588 with GStreamer / MIPI pipelines, concurrent processing, profiling, and on-site calibration.

Education

Jiujiang University *Jiujiang, China*
IoT Engineering *Sep. 2022 - Jun. 2026*

- **Core Coursework**: Data Structures, Computer Networks, Operating Systems, Computer Organization, Advanced Programming.
- **Language Certifications**: CET-4 and CET-6.

Research Experience

JJU CVI Lab
Computer Vision Research *Dec. 2022 - Sep. 2025*

- **Research Topics**: object detection, medical image segmentation, and edge AI; covered problem formulation, experimental hypothesis design, ablation studies, error analysis, and result synthesis.
- Built a biomedical imaging workflow for chromosome preprocessing, candidate-region detection, result review, and abnormal-sample annotation; integrated batch loading, manual correction, and result export in PyQt5.

Research Interests

`Core Research Interests`: My work focuses on **computer vision**, **edge AI**, and **robot learning**, spanning model training, architecture optimization, compression and quantization, and deployment on Jetson, RV1126B, and RK3588 platforms.

`Technical Focus`: Hands-on experience with real-robot data collection in LeRobot, visual-tactile multimodal training and inference, multi-camera video ingestion, concurrent processing, performance profiling, and hardware integration.

Publications

[1] TY-YOLO: An Attention-Based Multi-Scale Complex Scene Fire Smoke Detection Algorithm [Under Review]
First Author · JCR Q2

- Led model selection, architecture optimization, loss-function tuning, and manuscript preparation; combined attention mechanisms with multi-scale feature fusion to address small-target smoke, occlusion, and false positives in complex backgrounds.

[2] FgFEU-Net: A Fine-Grained Feature Extraction U-Net Model for Breast Tumor Segmentation Based on Transfer Learning [Paper]
Medical Imaging · Published

- Developed a transfer-learning U-Net with a VGG16 encoder, same-level feature fusion, and Combo Loss to address blurred tumor boundaries and foreground/background imbalance in breast ultrasound images.

[3] The Real-Time Intelligent Transportation Detection System Based on Edge-Cloud Collaborative Computing [Paper]
Real-Time Detection · Published

- Deployed YOLOv8n at the edge for vehicle recognition, traffic-flow statistics, congestion detection, wrong-way detection, and boundary-crossing alerts; used edge-cloud collaboration to reduce centralized processing latency.

Project Portfolio

VTLA *LeRobot · PI0.5 · Piper*

VLA / Robot Learning

- Extended LeRobot 0.5.2 with Piper leader-follower control, LTeach tactile sensing, and LeRobot-compatible recording, training, and inference workflows while preserving the upstream CLI/API.
- Implemented dedicated tactile pathways for PI0.5 and ACT-based DreamTacVLA, including lightweight tactile-token encoding, force-matrix conditioning, gradient accumulation, and regression-tested legacy HDF5 conversion.

OpenWorldTactile *Isaac Lab · UIPC · Force Fields*

Tactile Simulation & Force Fields

- Developed a UIPC-based compliant tactile membrane in Isaac Sim/Lab and reconstructed tactile-local 3D Fx/Fy/Fz fields from mesh deformation, including force projection, sign calibration, pressure, and shear estimation.
- Built an iterative validation suite covering contact/no-contact, membrane alignment, grasping, lifting, insertion, and lateral shear, while isolating incompatible Isaac Lab 2.1/2.3 environments and external asset dependencies.

OpenFireAlert *YOLO11 · TensorRT · Jetson*

Wildfire Monitoring & Alerting

- Trained a YOLO11 fire/smoke detector on D-Fire for 200 epochs, achieving 0.771 mAP@50 and 0.501 mAP@50-95; exported PyTorch, ONNX, and TensorRT artifacts for Jetson deployment.
- Engineered an end-to-end alert pipeline in which Jetson inference publishes annotated RTSP video and detection state to a Spring Boot control plane that fuses ESP32 smoke signals, suppresses duplicate alarms, and coordinates buzzer, email, and dashboard records.

TactileFlowField *PyTorch · RKNN · RV1126B*

Tactile Dense Displacement

- Developed a three-stage dual-frame network family for dense tactile displacement estimation, standardizing 15 model variants and 27 versioned training experiments behind strict input, checkpoint, and final-flow contracts.
- Built a PyTorch-to-ONNX-to-RKNN deployment path for RV1126B, redesigning later variants around RKNN operator constraints and supporting FP16/PTQ-INT8 export, per-camera baseline calibration, and multi-device inference.

EdgeVisTrack-RKNN *RKNN · Cython · RV1126B*

Multi-Camera Edge Tracking

- Designed a circular-marker tracking pipeline with regional RGB/Hough initialization, local template correlation, subpixel refinement, and loss recovery, using the CPU implementation as the behavioral reference.
- After profiling five-camera CPU execution at 98%–99% utilization on RV1126B, moved the hot path to batched RKNN backends and Cython/RKNN C APIs with FP16 LUT conversion, reusable buffers, and stage-level profiling.

Vision Workbench *PySide6 · OpenCV · Release Engineering*

Computer Vision Workbench

- Built a unified PySide6 desktop application exposing seven computer-vision workflows through modular Python APIs, from classical image processing and panorama reconstruction to YOLO detection, segmentation, and training.
- Productized the application with optional dependency isolation, 184 automated test functions, cross-platform Qt/accessibility checks, and verified wheel/Windows EXE updates using version gates, SHA-256 validation, dependency fingerprints, and rollback.

AdaptiveUI-SKILL *Python · Static Analysis · Agent Skills*

Responsive UI Audit & Repair

- Designed two explicit-invocation Agent Skills that turn responsive UI work into an audit-repair-verification workflow across HTML/CSS, React, Vue, Svelte, and Tailwind projects.
- Implemented a zero-dependency Python analyzer with 24 stable rules, severity/confidence metadata, suppressions, JSON Schema output, and CI thresholds, backed by 79 automated tests.

AgentTools *PowerShell · Diagnostics · Codex*

Agent Engineering & Diagnostics

- Built a read-only Codex diagnostics tool that correlates PowerShell profiles, Conda hooks, Python encoding, Git line endings, proxy/TUN state, and reconnect logs while redacting credentials and avoiding system mutation.
- Added cross-platform AGENTS.md templates, scoped task-context handoff, and runtime Skill inventory under an explicit-invocation plugin contract.

Skills

Vision	PyTorch, YOLO11, U-Net, OpenCV, Cython
Robotics	LeRobot, ACT, PI0, PI0.5, Piper, Isaac Sim, Isaac Lab, UIPC, Tactile Sensing
Platforms	Jetson, RV1126B, RK3588, GStreamer, RTSP, MIPI, ESP32
Deployment	ONNX, TensorRT, RKNN, RKNNLite, FP16, PTQ INT8, QAT
Engineering	Python, PyQt5, PySide6, Spring Boot 3, HDF5, Benchmarking, Logging, Profiling